Line Parameters II

Turn unequal tower geometry into one balanced per-phase inductance and reactance

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Lecture outline

1. Part 1: Three-phase configuration.
2. Part 2: Geometric mean radius (GMR) and phase inductance.
3. Part 3: Transposition and reactance.
4. Part 4: Exercises.

Establish the physical model, calculate its parameters, then solve complete exercises.

PART 1

Three-phase configuration

Three-phase conductor arrangements

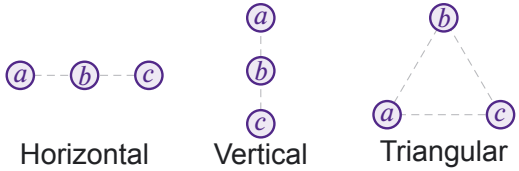
Configuration describes the **physical positions and spacing** of the three phases.



Near Vacaville, California.
Left: one circuit. Right: two circuits.

Photo: Hydrogen Iodide / Wikimedia Commons, CC BY-SA 3.0; cropped.

Common arrangements (cross-section)

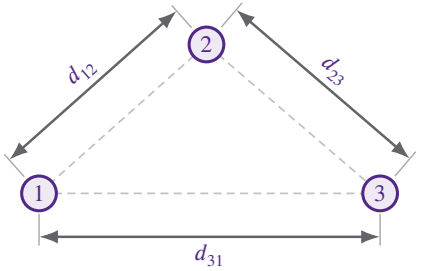


One circuit; one dot per phase position.
Triangular does not imply equal spacing.

Phase spacing changes flux linkage, so the same conductors can have different inductance and reactance.

Tower positions and phase labels

Fixed positions and center-to-center spacings



Example phase assignment

Position	1	2	3
Phase	A	B	C

For this assignment, the phase-pair distances are

$$D_{AB} = d_{12}, \qquad D_{BC} = d_{23},$$

$$D_{CA} = d_{31}.$$

Transposition changes phase assignments while the physical spacings remain fixed.

Three-phase current return

Balanced phase currents

$$I_a + I_b + I_c = 0$$

$$I_b + I_c = -I_a$$

The magnetic linkage of phase A depends on its distances to both return phases.

Model and output quantities

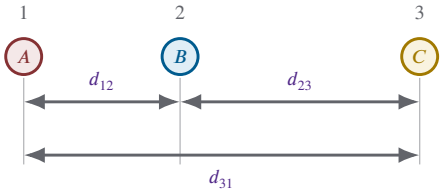
One identical, unbundled conductor per phase.

L'_ϕ: inductance per phase per unit length.

X'_ϕ: reactance per phase per unit length.

Including both return phases gives a per-phase result without a two-wire loop multiplier.

Unequal spacing



Distances to the other phases

Phase	Mutual distances
A	$d_{12},\ d_{31}$
B	$d_{12},\ d_{23}$
C	$d_{23},\ d_{31}$

Each phase has a different pair of distances in this arrangement.

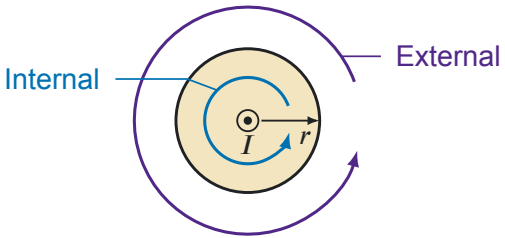
Balanced currents alone do not guarantee equal phase inductances.

PART 2

GMR and phase inductance

Conductor data for inductance

A conductor's inductance includes magnetic flux linkage both inside and outside its surface.



Cross-section; current out of the page
Schematic magnetic field lines

Physical radius r
Locates the conductor surface, where the external-flux calculation begins.

Geometric mean radius (GMR), D_s
An equivalent conductor radius that also accounts for the internal magnetic contribution.

GMR supplies the conductor's self-distance in the logarithmic inductance formula.

Magnetic self-distance

Consider a solid nonmagnetic conductor with uniform current and an equal return conductor at spacing $D \gg r$.

$$L'_{\text{cond}} \approx 2 \times 10^{-7} \left[\underbrace{\frac{1}{4}}_{\text{internal term}} + \underbrace{\ln \frac{D}{r}}_{\text{external term}} \right] \text{H/m}.$$

Choose D_s so one logarithm represents both contributions:

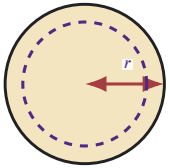
$$\ln \frac{D}{D_s} = \frac{1}{4} + \ln \frac{D}{r} \qquad \Longrightarrow \qquad \boxed{D_s = re^{-1/4} \approx 0.7788r}.$$

The same result becomes $L'_{\text{cond}} \approx 2 \times 10^{-7} \ln(D/D_s)$ H/m, with internal linkage included.

Source: Glover et al., sixth edition (2017), Sec. 4.4.

Conductor GMR

Ideal solid round conductor



$D_s = 0.7788r$

Real stranded conductor

use the supplied D_s.

Strand layout and current distribution affect the magnetic self-distance.

The supplied value already represents the conductor's self contribution.

Apply the 0.7788 factor to an ideal solid conductor radius only; use catalog GMR directly.

Three-phase flux linkage

For equal phase spacing D, choose the same distant reference D_0 for all three field contributions:

$$\lambda'_a = 2\times 10^{-7}\left[\underbrace{I_a \ln\frac{D_0}{D_s}}_{\text{own current, including internal flux}} + \underbrace{(I_b + I_c)\ln\frac{D_0}{D}}_{\text{fields of the other phases}}\right].$$

Balanced currents give $I_b + I_c = -I_a$, so the common reference cancels:

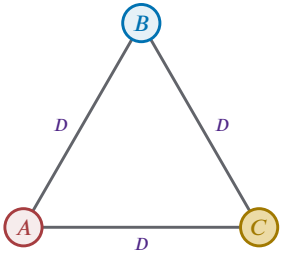
$$\lambda'_a = 2\times 10^{-7} I_a \left[\ln\frac{D_0}{D_s} - \ln\frac{D_0}{D}\right] = 2\times 10^{-7} I_a \ln\frac{D}{D_s}.$$

Divide this phase linkage by I_a to obtain the phase inductance per unit length.

Source: Glover et al., sixth edition (2017), Secs. 4.4–4.5.

Equal spacing

Balanced currents; identical unbundled conductors



Divide phase linkage by phase current:

$$L'_\phi = \frac{\lambda'_a}{I_a},$$

$$\boxed{L'_\phi = 2 \times 10^{-7} \ln\left(\frac{D}{D_s}\right) \text{ H/m.}}$$

Mutual geometric mean distance (GMD): $D_m = D$.

Equal spacing gives identical phase inductances without transposition.

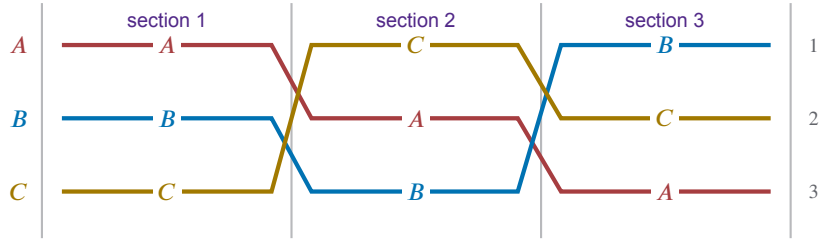
PART 3

Transposition and reactance

Conductor transposition

Equal spacing gives identical phase geometry. Unequal spacing makes magnetic coupling depend on phase position.

Transposition exchanges these positions along the route:



schematic position tracks; crossings show phase reassignment

To equalize average coupling, each phase must occupy every position for the same route length.

Complete transposition

Section k has length ℓ_k and phase-pair spacing $D_{AB}^{(k)}$.

Section k	Position 1	Position 2	Position 3	$D_{AB}^{(k)}$
1	A	B	C	d_{12}
2	C	A	B	d_{23}
3	B	C	A	d_{31}

Each phase occupies every position for the same route length:

$$\ell_1 = \ell_2 = \ell_3 = \frac{\ell}{3}$$

Equal section lengths give each spacing a weight of $1/3$ in the average linkage.

Geometric mean distance

Average the mutual logarithmic coefficient over three equal sections:

$$\frac{1}{3}\left(\ln\frac{D_0}{d_{12}}+\ln\frac{D_0}{d_{23}}+\ln\frac{D_0}{d_{31}}\right)=\ln\frac{D_0}{D_m},\qquad \boxed{D_m=(d_{12}d_{23}d_{31})^{1/3}}.$$

With balanced currents, $I_b+I_c=-I_a$:

$$\overline{\lambda'_a}=2\times10^{-7}\left[I_a\ln\frac{D_0}{D_s}+(I_b+I_c)\ln\frac{D_0}{D_m}\right]=2\times10^{-7}I_a\ln\frac{D_m}{D_s}.$$

Complete transposition gives all three phases the same average mutual geometry.

Phase inductance rate

For balanced currents, identical unbundled conductors, and equal spacing or complete transposition:

$$L'_\phi = 2 \times 10^{-7} \ln \frac{D_m}{D_s} \text{ H/m}$$

Mutual geometry: D_m

From the phase-center spacings.

Increasing D_m increases L'_ϕ.

Magnetic self data: D_s

From the conductor data.

Increasing D_s decreases L'_ϕ.

With fixed conductor data, route length and frequency do not enter L'_ϕ.

Source: Glover et al., sixth edition (2017), Secs. 4.5–4.6.

Reactance rate and total

Let f be frequency and ℓ the total route length.

Convert inductance to reactance

$$\omega = 2\pi f$$

$$X'_\phi = \omega L'_\phi$$

Use H/km for L'_ϕ to obtain Ω/km.
Convert mH to H first.

Convert rates to route totals

$$L_\phi = L'_\phi \ell$$

$$X_\phi = X'_\phi \ell = \omega L_\phi$$

Match the length unit in ℓ to the rate denominator.

> **For fixed conductor parameters, doubling frequency or route length doubles total reactance.**

PART 4

Exercises

Exercise 1: given and find

A fully transposed line has phase spacings $4, 4, 8$ m, GMR $D_s = 0.0100$ m, $f = 60$ Hz, and length 20.0 km.

1. Find D_m and the per-phase inductance per kilometre.
2. Calculate total per-phase series reactance.
3. Explain why this reactance is not multiplied by two.

Exercise 1: solution

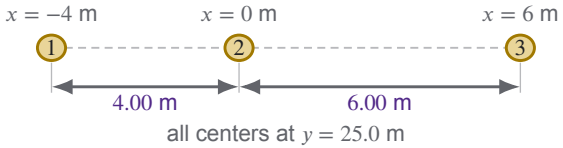
$$D_m = (4 \cdot 4 \cdot 8)^{1/3} = 5.03968 \text{ m}$$

$$L'_\phi = 2 \times 10^{-7} \ln \frac{5.03968}{0.0100} \approx 1.24450 \times 10^{-6} \text{ H/m} = 1.24450 \text{ mH/km}.$$

$$X_\phi = 2\pi(60)(1.24450 \times 10^{-3})(20.0) \approx 9.3833 \ \Omega$$

The balanced return currents are already included in the per-phase inductance formula.

Exercise 2: given



$$
\begin{aligned}
f &= 60 \text{ Hz}, \\
\ell &= 75.0 \text{ mi} \\
&= 120.7008 \text{ km}, \\
D_s &= 10.90 \text{ mm} \\
&= 0.01090 \text{ m}.
\end{aligned}
$$

Model assumptions: One identical, unbundled conductor per phase. Three equal transposition sections. Earth return and shield wires are omitted.

Exercise 2: find

(a) Calculate the three position spacings and transposed GMD D_m.

(b) Use supplied GMR to find L'_ϕ and X'_ϕ per kilometre.

(c) Find route L_ϕ and X_ϕ, then verify X_ϕ using miles.

(d) Quantify the error from using an arithmetic-mean spacing or substituting $r = 14.0$ mm for the supplied GMR.

The radius in (d) is a comparison input only. The stated GMR remains $D_s = 10.90$ mm.

Ex. 2(a): position spacings

The fixed-position coordinates are (x_i, y_i), in metres:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}.$$

Here $y_1 = y_2 = y_3 = 25.0$ m, so $d_{ij} = |x_i - x_j|$.

$$d_{12} = |-4 - 0| = 4.00 \text{ m},$$
$$d_{23} = |0 - 6| = 6.00 \text{ m},$$
$$d_{31} = |6 - (-4)| = 10.00 \text{ m}.$$

Include the outer spacing d_{31} when forming the three-distance product.

Ex. 2(a): GMD

Phase-pair distances by section (m)

Pair	$k=1$	$k=2$	$k=3$
AB	4	6	10
BC	6	10	4
CA	10	4	6

Every row has product $240\ \text{m}^3$.

Common geometric mean

$$D_m = (240\ \text{m}^3)^{1/3}$$

$$\boxed{D_m \approx 6.214465\ \text{m}}$$

Check both units and bounds: D_m is a length and lies between 4 m and 10 m.

Ex. 2(b): magnetic ratio

Convert the supplied GMR to the same unit as D_m:

$$D_s = 10.90\ \text{mm} = 0.01090\ \text{m}$$

$$\frac{D_m}{D_s} \approx \frac{6.214465\ \text{m}}{0.01090\ \text{m}} \approx 570.1344$$

$$\boxed{\ln\left(\frac{D_m}{D_s}\right) \approx 6.345872}$$

The logarithm requires a dimensionless ratio. Here $D_m > D_s$, so the result is positive.

Ex. 2(b): parameter rates

$$L'_\phi = 2\times10^{-7}(6.345872) \approx 1.269174\times10^{-6}\ \text{H/m} = 1.269174\ \text{mH/km}.$$

At 60 Hz, convert mH/km to H/km before multiplying by $2\pi f$:

$$X'_\phi = 2\pi(60)(1.269174\times10^{-3}) \approx \boxed{0.4784675\ \Omega/\text{km}}.$$

These per-phase rates apply to any route length with the same geometry, conductor, and frequency.

Ex. 2(c): route totals

Use $\ell = 120.7008$ km and retain unrounded rates in the calculation.

Inductance

$$\begin{aligned} L_\phi &= L'_\phi \ell \\ &\approx (1.269174 \times 10^{-3}\ \text{H/km}) \\ &\quad \times 120.7008\ \text{km} \\ &\approx \boxed{0.153190\ \text{H}}. \end{aligned}$$

Reactance

$$\begin{aligned} X_\phi &= X'_\phi \ell \\ &\approx (0.4784675\ \Omega\text{/km}) \\ &\quad \times 120.7008\ \text{km} \\ &\approx \boxed{57.7514\ \Omega}. \end{aligned}$$

Independent check: $2\pi(60)(0.153190) \approx 57.751\ \Omega$ **per phase.**

Ex. 2(d): comparisons

The mile-based calculation confirms the unit conversion:

$$(0.7700188\ \Omega/\text{mi})(75.0\ \text{mi}) \approx 57.7514\ \Omega$$

Inputs	X_ϕ (Ω)	Relative error
GMD and supplied GMR	57.7514	reference
Arithmetic mean 6.6667 m	58.3906	+1.107%
Radius 14.0 mm substituted for GMR	55.4736	−3.944%

Matching results in miles and kilometres cannot detect an incorrect geometry or conductor input.

Summary

For balanced currents and identical unbundled conductors, use equal spacing or complete transposition.

Choose the two distance inputs

D_m: geometric mean of the three position spacings.

D_s: supplied conductor GMR, or the ideal-solid value when that model applies.

Calculate the per-phase result

$$L'_\phi = 2 \times 10^{-7} \ln(D_m / D_s)$$

$$X_\phi = 2\pi f L'_\phi \ell$$

Use the same unit for D_m, D_s; the coefficient gives H/m, so use ℓ in metres.

Validate the units, the equal-spacing limit, and the expected response to changes in spacing and GMR.